ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 44	$\frac{27\times57+39\times35}{66}=44.$
2	**(a)** 33	$\frac{14\times31+61}{15}=33.$
3	**(b)** 69	$\frac{70\times6+65\times6+75\times3}{15}=69.$
4	**(d)** 27%	$\frac{60\times15+120\times33}{180}=27\%.$
5	**(a)** 153	17% of $900=\frac{17}{100}\times900=153.$
6	**(d)** $\frac{w_1x_1+w_2x_2}{w_1+w_2}$	Formula.
7	**(d)** 30	Sum of the extra numbers $=46\times45-30\times54=450$; divide by 15.
8	**(c)** 31	$\frac{8\times32-39}{7}=31.$
9	**(c)** 54	$\frac{35\times65+55\times47}{90}=54.$
10	**(c)** 32	$\frac{21\times31+53}{22}=32.$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total of first group $=27\times67=1809$; Total of second group $=45\times35=1575$ (1); Combined average $=\frac{3384}{72}=47$ (1).	1+1
12	Weighted sum $=90\times6+75\times3+85\times3=1020$; Sum of weights $=12$ (1); Weighted average $=85$ (1).	1+1
13	Books $=493$; Sports $=187$ (1); Remaining share 60%, amount $=1020$ (1).	1+1
14	Total of first group $=30\times53=1590$; Total of second group $=27\times72=1944$ (1); Combined average $=\frac{3534}{57}=62$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pure part in the first $=110\times\frac{38}{100}=41.8$ mL; Pure part in the second $=96.8$ mL (1); Total $=138.6$ mL in 330 mL (1); Concentration $=42\%$ (1).	1+1+1
16	Sum of 10 observations $=10\times53=530$ (1); New sum $=530+64=594$ (1); New mean $=\frac{594}{11}=54$ (1).	1+1+1
17	Total score of all $=56\times54=3024$; Total score of the first group $=15\times69=1035$ (1); Total score of the additional group $=1989$ (1); Average $=\frac{1989}{39}=51$ (1).	1+1+1
	OR: Pure part in the first $=390\times\frac{41}{100}=159.9$ mL; Pure part in the second $=16.9$ mL (1); Total $=176.8$ mL in 520 mL (1); Concentration $=34\%$ (1).	1+1+1
18	Sum of 27 observations $=27\times30=810$ (1); New sum $=810+58=868$ (1); New mean $=\frac{868}{28}=31$ (1).	1+1+1
	OR: Total score of all $=55\times42=2310$; Total score of the first group $=24\times58=1392$ (1); Total score of the additional group $=918$ (1); Average $=\frac{918}{18}=51$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Totals: 1320, 1222, 1672 (1); Grand total $=4214$ for 86 students (2); Average $=49$ (2).	1+2+2
	OR: January food $=1539$ (1); February food $=1960$ (1); February is higher by 421 (1); The larger total outweighs the smaller share in February (2).	1+1+1+2
20	Totals: 1720, 3432, 1456 (1); Grand total $=6608$ for 112 students (2); Average $=59$ (2).	1+2+2

	OR: January food $= 1127$ (1); February food $= 1116$ (1); January is higher by 11 (1); The larger share outweighs the smaller total in January (2).	1+1+1+2

Mistakes to avoid:

- Taking the combined average as $\frac{a_1 + a_2}{2}$ when the groups have different sizes.
- Adding the percentages of two solutions to find the mixture, or averaging the two concentrations without using the volumes.
- Dividing the sum of weighted scores by the number of items instead of by the sum of the weights.
- Forgetting to change the volume when water is added: the pure amount stays the same but the total increases.
- Using a percentage as if it were an amount (e.g. 20% of 5 L is 1 L, not 20 L).
- Reading a 100% stacked bar as if the bar lengths showed the actual totals; the actual totals are not shown.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 71	$\frac{93\times3+80\times2+57\times6}{11}=71.$
2	**(a)** 23%	$\frac{260\times33+200\times10}{460}=23\%.$
3	**(a)** 285	25% of $1140=\frac{25}{100}\times1140=285.$
4	**(d)** between the two concentrations	It is a weighted average.
5	**(d)** 67	Sum of the extra numbers $=47\times42-30\times39=804$; divide by 12.
6	**(b)** 54	$\frac{15\times56-84}{14}=54.$
7	**(d)** 43	$\frac{58\times46+29\times37}{87}=43.$
8	**(a)** 58	$\frac{9\times60+40}{10}=58.$
9	**(c)** 64	$\frac{50\times6+68\times3+88\times3}{12}=64.$
10	**(a)** 23%	$\frac{80\times10+260\times27}{340}=23\%.$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Weighted sum $=89\times2+81\times5+61\times5=888$; Sum of weights $=12$ (1); Weighted average $=74$ (1).	1+1
12	Books $=756$; Sports $=297$ (1); Remaining share 61%, amount $=1647$ (1).	1+1
13	Total of first group $=36\times60=2160$; Total of second group $=52\times38=1976$ (1); Combined average $=\frac{4136}{88}=47$ (1).	1+1
14	Weighted sum $=58\times4+67\times4+66\times6=896$; Sum of weights $=14$ (1); Weighted average $=64$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pure part in the first $=320\times\frac{32}{100}=102.4$ mL; Pure part in the second $=164$ mL (1); Total $=266.4$ mL in 720 mL (1); Concentration $=37\%$ (1).	1+1+1
16	Sum of 16 observations $=16\times40=640$ (1); New sum $=640+57=697$ (1); New mean $=\frac{697}{17}=41$ (1).	1+1+1
17	Total score of all $=43\times48=2064$; Total score of the first group $=32\times47=1504$ (1); Total score of the additional group $=560$ (1); Average $=\frac{560}{16}=35$ (1).	1+1+1
	OR: Pure part in the first $=420\times\frac{18}{100}=75.6$ mL; Pure part in the second $=141.9$ mL (1); Total $=217.5$ mL in 750 mL (1); Concentration $=29\%$ (1).	1+1+1
18	Sum $=11\times46=506$ (1); Remaining sum $=506-76=430$ (1); New mean $=\frac{430}{10}=43$ (1).	1+1+1
	OR: Total score of all $=31\times48=1488$; Total score of the first group $=12\times52=624$ (1); Total score of the additional group $=864$ (1); Average $=\frac{864}{36}=24$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Totals: 2190, 1820, 1890 (1); Grand total $=5900$ for 100 students (2); Average $=59$ (2).	1+2+2
	OR: January food $=940$ (1); February food $=725$ (1); January is higher by 215 (1); The larger share outweighs the smaller total in January (2).	1+1+1+2
20	Totals: 1344, 1189, 624 (1); Grand total $=3157$ for 77 students (2); Average $=41$ (2).	1+2+2
	OR: January food $=540$ (1); February food $=462$ (1); January is higher by 78 (1); The larger share outweighs the smaller total in January (2).	1+1+1+2

Mistakes to avoid:

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- Dividing the sum of weighted scores by the number of items instead of by the sum of the weights.
- Forgetting to change the volume when water is added: the pure amount stays the same but the total increases.
- Using a percentage as if it were an amount (e.g. 20% of 5 L is 1 L, not 20 L).
- Reading a 100% stacked bar as if the bar lengths showed the actual totals; the actual totals are not shown.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 330	33% of $1000 = \frac{33}{100} \times 1000 = 330$.
2	**(a)** total distance divided by total time	Definition of average speed.
3	**(a)** 69	Sum of the extra numbers $= 60 \times 32 - 8 \times 33 = 1656$; divide by 24.
4	**(d)** 35	$\frac{15 \times 37 - 65}{14} = 35$.
5	**(c)** 54	$\frac{38 \times 76 + 44 \times 35}{82} = 54$.
6	**(d)** 47	$\frac{12 \times 46 + 59}{13} = 47$.
7	**(c)** 69	$\frac{66 \times 2 + 78 \times 4 + 63 \times 5}{11} = 69$.
8	**(a)** 26%	$\frac{180 \times 25 + 90 \times 28}{270} = 26\%$.
9	**(c)** 304	38% of $800 = \frac{38}{100} \times 800 = 304$.
10	**(a)** increases	A bigger value pulls the mean up.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Books $= 378$; Sports $= 196$ (1); Remaining share 59%, amount $= 826$ (1).	1+1
12	Total of first group $= 46 \times 77 = 3542$; Total of second group $= 30 \times 39 = 1170$ (1); Combined average $= \frac{4712}{76} = 62$ (1).	1+1
13	Weighted sum $= 82 \times 5 + 45 \times 3 + 73 \times 5 = 910$; Sum of weights $= 13$ (1); Weighted average $= 70$ (1).	1+1
14	Books $= 351$; Sports $= 99$ (1); Remaining share 50%, amount $= 450$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pure part in the first $= 130 \times \frac{36}{100} = 46.8$ mL; Pure part in the second $= 39$ mL (1); Total $= 85.8$ mL in 390 mL (1); Concentration $= 22\%$ (1).	1+1+1
16	Sum $= 19 \times 40 = 760$ (1); Remaining sum $= 760 - 94 = 666$ (1); New mean $= \frac{666}{18} = 37$ (1).	1+1+1
17	Total score of all $= 37 \times 62 = 2294$; Total score of the first group $= 30 \times 53 = 1590$ (1); Total score of the additional group $= 704$ (1); Average $= \frac{704}{32} = 22$ (1).	1+1+1
	OR: Pure part in the first $= 180 \times \frac{27}{100} = 48.6$ mL; Pure part in the second $= 86.4$ mL (1); Total $= 135$ mL in 450 mL (1); Concentration $= 30\%$ (1).	1+1+1
18	Sum $= 11 \times 59 = 649$ (1); Remaining sum $= 649 - 49 = 600$ (1); New mean $= \frac{600}{10} = 60$ (1).	1+1+1
	OR: Total score of all $= 51 \times 45 = 2295$; Total score of the first group $= 27 \times 35 = 945$ (1); Total score of the additional group $= 1350$ (1); Average $= \frac{1350}{18} = 75$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Totals: $1344, 1980, 861$ (1); Grand total $= 4185$ for 93 students (2); Average $= 45$ (2).	1+2+2
	OR: January food $= 1188$ (1); February food $= 1248$ (1); February is higher by 60 (1); The larger total outweighs the smaller share in February (2).	1+1+1+2
20	Totals: $1876, 1152, 1220$ (1); Grand total $= 4248$ for 72 students (2); Average $= 59$ (2).	1+2+2
	OR: January food $= 1080$ (1); February food $= 630$ (1); January is higher by 450 (1); The larger share outweighs the smaller total in January (2).	1+1+1+2

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